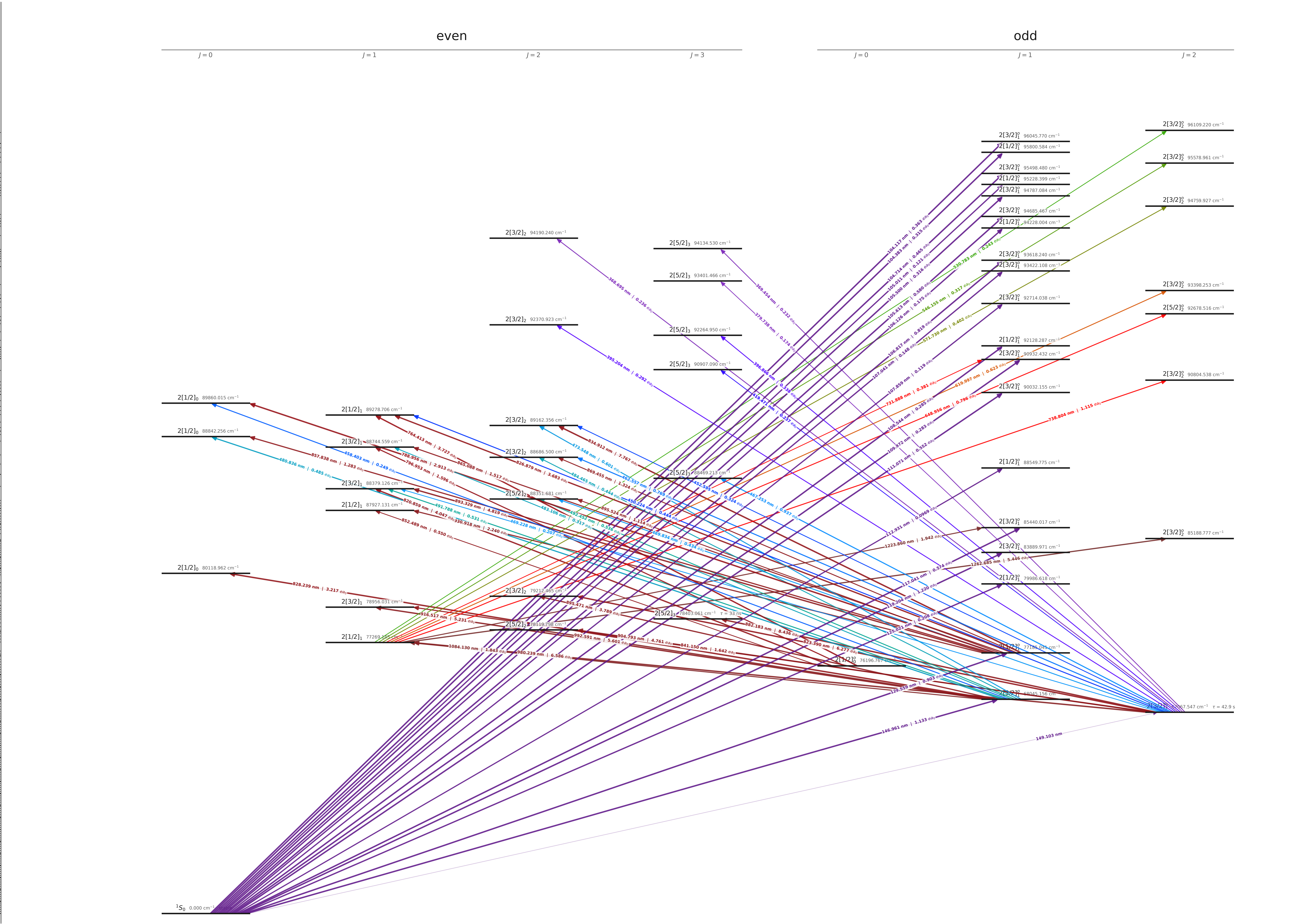


Xenon

Xe (Xe I). Strongest classified transitions below 2 μm (NIST): 51 levels, 71 lines.

Xe⁺ ionisation limit 97833.79 cm⁻¹ = 12.12984 eV (102.214 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er J' (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
¹ S ₀ - 2[3/2] ₂ ^o	149.1034	-	2010.6345	-	2.331e-02	3.71e-09	100	149.1 nm M2 decay of the metastable 3P2 state
2[1/2] ₂ ^o - 2[3/2] ₁	820.8591	820.6335	365.2179	4.047	2.000e+07	-	-	820.86 nm
2[3/2] ₂ ^o - 2[3/2] ₂	823.3897	823.1634	364.0955	6.277	2.860e+07	-	-	823.2 nm
2[3/2] ₂ ^o - 2[3/2] ₁	841.1501	840.9190	356.4078	1.642	3.060e+06	-	-	841.15 nm
2[3/2] ₂ ^o - 2[5/2] ₃	882.1832	881.9410	339.8302	8.436	3.000e+07	4.818	99.1	881.9 nm

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2τ_u).

¹²⁹Xe hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ₂ ^o	-2384.4(12)	-	F=3/2: +3576.60 F=5/2: -2384.40
2[3/2] ₂ ^o	-959.1(7)	-	F=1/2: +959.10 F=3/2: -479.55
2[1/2] ₂ ^o	-5808.0(20)	-	F=1/2: +5808.00 F=3/2: -2904.00
2[3/2] ₁	-1308.9(6)	-	F=1/2: +1308.89 F=3/2: -654.45
2[3/2] ₂	-889.6(4)	-	F=3/2: +1334.40 F=5/2: -889.60
2[3/2] ₁	-4424.0(10)	-	F=1/2: +4424.00 F=3/2: -2212.00
2[3/2] ₂	-2895(5)	-	F=3/2: +4341.90 F=5/2: -2894.60
2[1/2] ₁	1977.0(10)	-	F=1/2: -1977.00 F=3/2: +988.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
2[1/2] ₂ ^o - 2[3/2] ₁	820.859	136-134	-80.9(20)
2[1/2] ₂ ^o - 2[3/2] ₁	820.859	136-132	-132.4(21)
2[1/2] ₂ ^o - 2[3/2] ₁	820.859	136-131	-203(4)
2[1/2] ₂ ^o - 2[3/2] ₁	820.859	136-130	-182(5)
2[1/2] ₂ ^o - 2[3/2] ₁	820.859	136-129	-240(3)
2[1/2] ₂ ^o - 2[3/2] ₁	820.859	136-128	-242.4(24)
2[3/2] ₂ ^o - 2[3/2] ₁	841.150	136-134	-.95(4)
2[3/2] ₂ ^o - 2[3/2] ₁	841.150	136-132	-147(4)
2[3/2] ₂ ^o - 2[3/2] ₁	841.150	136-131	-222(7)
2[3/2] ₂ ^o - 2[3/2] ₁	841.150	136-130	-200(4)
2[3/2] ₂ ^o - 2[3/2] ₁	841.150	136-129	-275(5)
2[3/2] ₂ ^o - 2[3/2] ₁	841.150	136-128	-261(5)

Shift = ν(first isotope) - ν(second isotope).